

Osteoporosis — Clinical One-Pager

Screen · diagnose by 3 criteria · bisphosphonate first-line · monitor & holiday. 2023 ACP living guideline. Adapted from TeachIM.org (CC BY-NC 4.0) — clinician-reviewed before use.

Diagnosis (any of 3)

- **DXA T-score ≤ -2.5**
- **Fragility fracture** — low-trauma fx of hip, vertebra, wrist, pelvis, ribs, humerus (dx at any T-score; treat without waiting for DXA)
- **High FRAX** — 10-yr $\geq 20\%$ major OR $\geq 3\%$ hip

Classification	T-score
Normal	> -1.0
Osteopenia	-1.0 to -2.5
Osteoporosis	< -2.5

Most fractures occur above a -2.5 T-score — use clinical risk, not BMD alone.

Screening — who

- All women ≥ 65 ; postmenopausal women < 65 with ≥ 1 risk factor (use FRAX); men with major risk factors.
- **Risk factors:** glucocorticoids (≥ 20 mg $\times 3$ mo), aromatase inhibitors, ADT, antiepileptics; tobacco, alcohol $> 2/d$; premature menopause, hypogonadism, hyperparathyroidism, T1DM, Cushing; celiac/IBD, RA, CLD, CKD, HIV, myeloma; low BMI, FHx hip fx, prior fx.

Secondary work-up

Indicated in **all men, premenopausal women**, or unexplained/ < 50 .

- **Initial:** CBC, CMP, Mg, Phos, 25-OH vit D, testosterone (men).
- **Directed:** TSH, intact PTH, celiac panel, SPEP/UPEP, AM cortisol, FSH/LH/estradiol.

Primary interventions (ALL)

- Calcium ~ 1200 mg/d (diet preferred); vitamin D 800 IU/d (50,000 IU/wk if < 40).
- Weight-bearing high-impact exercise; smoking cessation; limit alcohol.
- **VITAL (NEJM 2022):** vit D alone did not cut fractures — but replete when starting a bisphosphonate; don't delay treatment.

Targeted therapy by risk

Agent	Role / notes
Bisphosphonate (alendronate PO wkly; zoledronic IV yrly)	1st-line , eGFR ≥ 35 . PO: empty stomach, 8 oz water, upright ≥ 30 min.
Denosumab (SQ q6 mo)	2nd-line (intol/eGFR < 35). Never stop without bridging to a bisphosphonate → rebound vert fx; do NOT follow with PTH analog.
Anabolic (teriparatide SQ qd; romosozumab SQ qmo)	Very-high-risk ($T \leq -3.5$, or ≤ -2.5 + fx). Anabolic FIRST, then antiresorptive. Avoid romo if recent MI/CVA.
SERM (raloxifene)	Rare; selected postmenopausal (also $\uparrow$ breast-ca risk). Harms: VTE, hot flashes.

Harms to counsel: esophagitis, hypocalcemia; rare ONJ & atypical femur fracture.

Monitoring & drug holiday

Situation	Repeat DXA
Low risk (normal)	~ 15 yr
Moderate (osteopenia)	1–2 yr
On IV bisphosphonate	~ 3 yr
On oral bisphosphonate	~ 5 yr
During drug holiday	2–4 yr

Drug holiday after 5 yr PO / 3 yr IV if BMD stable/improved & risk fallen. If worse / still high-risk → continue or refer endo (denosumab/anabolic).

2023–24 updates

- ACP: bisphosphonate 1st-line (strong in women); denosumab 2nd-line.
- Very-high-risk women: anabolic-first then antiresorptive (greater/earlier benefit).
- ASBMR/BHOF 2024: treat-to-target (goal BMD), not fixed duration.